

Adaptive Topological Similarity Learning for Incomplete Multi-view Unsupervised Feature Selection

Lin Xu, Ke Li, Dongjie Wang, Fanyin Zhou, Fengmao Lv, Tianrui Li, *Senior Member, IEEE* and Yanyong Huang*

Abstract—Although multi-view unsupervised feature selection is a promising technique for dimensionality reduction on unlabeled multi-view data, existing methods cannot directly address incomplete data, where certain samples are missing in specific views. These methods typically begin by imputing missing data using predetermined values, followed by performing feature selection on the completed dataset. However, the separation of imputation and feature selection processes fails to exploit their inherent synergy, as local structural information obtained from feature selection could guide the imputation process and, in turn, improve the overall effectiveness of feature selection. In addition, previous methods rely on similarity graphs based on Euclidean distance to preserve the local manifold structure but overlook the topological relationships within the data, thereby hindering accurate capture of intrinsic structures. In this paper, we propose an adaptive topological similarity learning for incomplete multi-view unsupervised feature selection method (ATSL-IMUFS) to address the aforementioned issues. ATSL-IMUFS first integrates multi-view feature selection and missing data imputation into a unified learning framework. Then, it adaptively learns similarity graphs for each view while simultaneously capturing the consensus topological relationship across views, effectively characterizing the local manifold structure. Extensive experiments conducted on real-world datasets demonstrate the superior performance of ATSL-IMUFS compared to competing methods.

Index Terms—Multi-view unsupervised feature selection, Incomplete data, Similarity graph, Consensus topological relationship.

I. INTRODUCTION

MULTI-VIEW data, where each sample is characterized by heterogeneous feature sets derived from different sources or perspectives, are common in real-world applications [1]. For example, in news classification, an article can be represented using textual information in different languages gathered from multiple sources, such as BBC, Deutsche Welle,

and Xinhua News [2]. In intelligent transportation systems, acoustic and seismic sensors are often used to capture diverse signals from the same vehicle, including engine noise, tire friction, and ground vibrations [3]. In these applications, data from different views are typically represented in a high-dimensional feature space that often includes redundant and noisy features. This can lead to the “curse of dimensionality” ultimately degrading the performance of clustering and classification tasks [1], [4]. Moreover, acquiring a large amount of labeled data remains a major challenge in practice, as it is both labor-intensive and time-consuming [5]. Therefore, reducing the dimensionality of multi-view unlabeled data to enhance the performance of subsequent tasks has become a critical problem in many practical applications.

Multi-view unsupervised feature selection (MUFS) offers an effective solution to the above problem by selecting a compact set of informative features from the original multi-view data [1], [6], [7]. In recent years, numerous MUFS methods have been proposed, which can generally be divided into two main categories. The first category involves simply concatenating the features from multiple views and then inputting them into traditional single-view feature selection methods. Typical single-view methods in this category include Laplacian Score (LS) [8], General Adaptive Unsupervised Feature Selection with Auto-weighting (GAWFS) [9], and unsupervised feature selection with High-order Similarity Learning (HSL) [10]. Although these methods can improve feature selection to some extent, they treat each view independently and fail to leverage the underlying correlations between different views. Instead of combining features from different views, the second category of MUFS methods focuses on constructing models directly from multi-view data to facilitate feature selection. Within this category, a typical approach is the Joint Multi-View Unsupervised Feature Selection and Graph learning method (JMVFG) [11]. It utilizes orthogonal decomposition to derive view-specific basis matrices and a consensus clustering indicator matrix, while learning a cross-view similarity graph to maintain the data’s local structure. Besides, Yuan et al. [12] have proposed the method named a tensor low-rank constrained graph embedding model for multi-view unsupervised feature selection (TLR-MUFS). While these proposed methods have demonstrated promising performance in feature selection, they rely on a common assumption that each instance exists in

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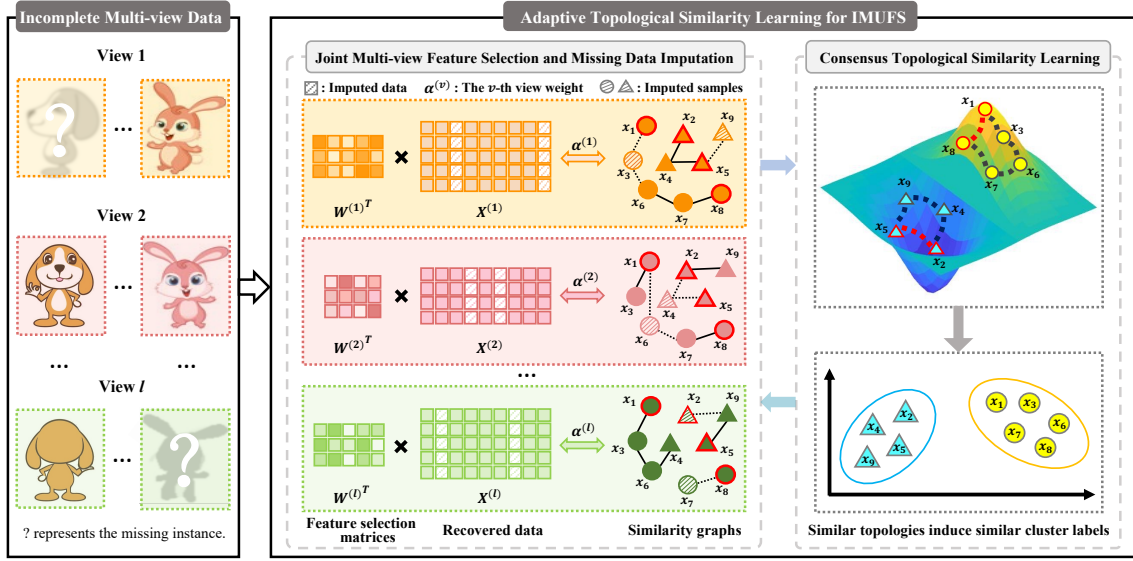


Fig. 1: The framework of the proposed ATSL-IMUFS.

all views, which is not always valid in real-world scenarios. For example, in tumor diagnosis, CT scans, MRI, and PET provide three distinct perspectives (views) for patients, though some may lack access to MRI or PET due to limited equipment availability [13]. As a result, the above-mentioned methods cannot be directly applied to incomplete multi-view data. There is still limited research on unsupervised feature selection for such data. Xu et al. [14] have introduced a Cross-View Feature Selection (CVFS) framework for incomplete multi-view data, which identifies informative features by utilizing cross-view interactions at both the instance and feature levels. Besides, Huang et al. [7] have proposed a method called Complementary and Consensus Learning-based Incomplete Multi-view Unsupervised Feature Selection (C^2 IMUFS). This approach incorporates complementary and consensus information across views into an extended weighted non-negative matrix factorization model to facilitate feature selection. However, these methods treat feature selection and data imputation as two independent steps, neglecting their potential synergy. Specifically, the local structures derived from feature selection can guide data imputation, while the refined imputed data can, in turn, enhance feature selection [15]. Consequently, several MUFS methods have recently been proposed to integrate feature selection and data imputation within a unified learning framework. TIME-FS [16] jointly performs adaptive missing-view imputation, discriminative feature selection, and tensor-based anchor graph learning to identify discriminative features from incomplete multi-view data. Yang et al. [17] combine self-representation learning, a low-rank tensor constraint, and sample diversity learning for simultaneous data imputation and feature selection. Nevertheless, these methods construct similarity graphs based on Euclidean distances to preserve the local data structure. However, such graphs cannot accurately capture the intrinsic relationships among samples residing on a nonlinear manifold. For example, two samples that appear far apart in Euclidean distance may still demonstrate strong topological consistency if they are connected through

a series of consecutive neighboring samples [18], [19]. By neglecting topological relationships, existing methods often fail to effectively preserve the local data structure, thereby degrading feature selection performance.

To address these issues, we propose a novel unsupervised feature selection method for incomplete multi-view data, called Adaptive Topological Similarity Learning for Incomplete Multi-view Unsupervised Feature Selection (ATSL-IMUFS). Specifically, we first integrate multi-view feature selection and missing data imputation into a unified learning framework by leveraging spectral graph theory. Subsequently, the similarity graph of each view and the consensus topological relationships across views are adaptively learned to capture the intrinsic similarity of the local manifold structure in multi-view data. By jointly exploiting intra-view and inter-view local structural information, the proposed module can more effectively guide feature selection and facilitate data imputation. Different from existing unified learning-based IMUFS methods that ignore the topological relationships within the data, we adaptively learn the consensus topological structure across views and utilize it to guide the construction of view-specific similarity graphs, thus more accurately preserving the local manifold structure and enhancing feature selection performance. Furthermore, we propose an alternating iterative optimization algorithm to solve the proposed problem, accompanied by a thorough analysis of its convergence and complexity. Extensive experiments on eight real-world datasets are conducted to validate the effectiveness of ATSL-IMUFS compared to competing methods. The overall framework of ATSL-IMUFS is shown in Fig. 1.

To sum up, the main contributions of this paper are summarized as follows: (i) To the best of our knowledge, we are the first to address the problems of multi-view unsupervised feature selection on incomplete data by leveraging consensus topological relationships across views and view-specific similarity graphs. (ii) We propose a unified learning framework induced by topological relationships, which adaptively

learns cross-view consistent topological structures to guide the construction of view-specific similarity graphs, and seamlessly integrates feature selection and missing data imputation. (iii) An efficient alternating optimization algorithm is developed and validated through comprehensive experiments on real-world multi-view datasets, demonstrating the superiority of the proposed method over other competing methods.

The rest of this paper is organized as follows. Section 2 offers a brief review of related work on MUFS. In Section 3, we formulate ATSL-IMUFS and provide an effective iterative optimization algorithm for the proposed problem, along with convergence and complexity analyses. Section 4 reports a series of experiments conducted to validate the effectiveness of the proposed ATSL-IMUFS. Finally, Section 5 concludes the paper.

II. RELATED WORK

In this section, we briefly introduce some representative works on MUFS, focusing on single-view-based and multi-view-based approaches.

A. Single-view Unsupervised Feature Selection Methods

Single-view unsupervised feature selection methods are roughly divided into three categories: filter-based, wrapper-based, and embedded-based approaches. The first type of methods evaluates the relevance of each feature individually using specific criteria and selects the most informative ones accordingly. A representative method Laplacian Score (LS) [8] measures feature importance based on its ability to preserve the local structure. Filter-based methods neglect the interdependencies among selected features and operate independently of specific learning algorithms, potentially resulting in suboptimal performance in downstream tasks. The second type of methods typically employs a specific learning algorithm to evaluate selected features iteratively until the optimal feature set is identified. Wen et al. [20] proposed an AdaBoost-guided feature selection approach that integrates Haar-like features and class labels to construct effective weak classifiers and identify discriminative visual features for vehicle detection. Kabir et al. [21] proposed a neural network-based feature selection method that incrementally incorporates features by exploiting their correlations, while dynamically training the neural network model. The trained model is then used to evaluate the quality of the selected feature subsets, and this process is repeated until the optimal subset is achieved. The primary drawback of wrapper-based methods is their high computational cost, which results from the process of identifying the features that yield the best performance. The last type of methods integrates feature selection into the learning phase of a specific algorithm, allowing for the efficient identification of important features with reduced computational cost. HSL [10] projects high-dimensional data onto a low-dimensional embedding space while maintaining both first-order and higher-order similarity structures, thereby enabling the identification of discriminative features. GAWFS [9] integrates non-negative matrix factorization with adaptive graph

learning to simultaneously perform clustering and select discriminative features through the learning of a feature weighting matrix. Zhou et al. [22] proposed a robust unsupervised feature selection method that replaces the conventional Frobenius-norm reconstruction error with the Maximum Correntropy Criterion, thereby effectively suppressing heavy-tailed non-Gaussian noise and outliers. When applying single-view-based unsupervised feature selection methods to multi-view data, they simply concatenate features from different views, overlooking the latent correlations between views, which ultimately leads to suboptimal performance.

B. Multi-view Unsupervised Feature Selection Methods

In contrast to the aforementioned single-view-based methods, multi-view unsupervised feature selection methods directly process multi-view data and utilize the underlying correlation information across views to improve performance. Hu et al. [23] proposed an unsupervised kernel-based multi-view feature selection approach (UKMFS) that combines robust self-representation with binary hashing. By applying kernel mapping to unify multi-view data representations, UKMFS constructs consistent graphs across views under low-rank constraints to reduce noise and utilizes binary hashing codes to guide feature selection. Xu et al. [24] utilized graph regularization to capture consensus information across views and kernel-based dependency measures to explore inter-view diversity, both of which were embedded into the feature selection process. TLR-MUFS [12] performs feature selection by exploiting intra-view structural information derived from view-specific graphs and enforcing cross-view consistency via tensor low-rank regularization. Zhou et al. [26] proposed a consistency-exclusivity guided unsupervised multi-view feature selection method, termed CE-UMFS. CE-UMFS leverages the Hilbert-Schmidt Independence Criterion to maintain inter-view exclusivity and employs nuclear norm regularization to enforce cross-view consistency, both of which contribute to the identification of discriminative features. JMVFG [11] decomposes the projected data matrix into view-specific basis matrices and a clustering indicator matrix that ensures cross-view consistency, while preserving the local data structure using consensus graph learning. Liang et al. [28] proposed a MUFS method called TRCA-CGL, which combines feature selection, consensus similarity graph learning, and tensor principal component analysis into a unified learning framework. Cao et al. [25] simultaneously leveraged neighbor information from multi-order similarity graphs and consistency information from a shared latent representation to identify informative features. Zhou et al. [29] introduced a structural regularization-based MUFS method that simultaneously learns view-specific and consensus similarity graphs via an automatic weighting strategy to preserve data locality and achieve feature selection. CDMIMFS [6] leverages fuzzy rough set and sparse learning to capture class-specific discriminability and extract multiscale information, thereby guiding multi-view feature selection. Cao et al. [27] employed spectral analysis on multiple Laplacian graphs to derive a consensus clustering indicator and heterogeneity similarity graphs, aiding feature selection by

TABLE I: Comparison of different multi-view unsupervised feature selection methods.

Comparisons	References	Ours	[8]	[21]	[10]	[9]	[23]	[24]	[25]	[26]	[27]	[28]	[29]	[11]	[12]	[14]	[7]	[17]	[30]	[16]
Multi-view Data Support		✓					✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Graph Learning		✓	✓		✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓		✓	✓	✓	✓
Incomplete Data Handling		✓														✓	✓	✓	✓	✓
Joint Feature Selection and Adaptive Imputation		✓															✓			✓
Topological Relationships		✓																		

utilizing both heterogeneous and homogeneous information across multi-view data. Cao et al. [31] employ a self-paced learning strategy to progressively select informative samples for training, and utilize tensor learning to capture higher-order view correlations and discriminative information for feature selection. Huang et al. [32] propose a heterogeneous regularization to capture complex correlations within spectral-spatial data, thus enabling the identification of representative features. Yang et al. [33] adaptively learn hypergraph Laplacian matrices within a self-paced learning framework to capture multi-view correlations and preserve local manifold structures, facilitating unsupervised feature selection. The previously mentioned MUFS methods typically assume that all samples are fully observed in each view, making them unsuitable for incomplete multi-view data, where some samples may be missing in certain views. In order to select features from incomplete multi-view data, CVFS [14] first transforms incomplete multi-view data into complete data by filling in missing values with the mean and then applies non-negative matrix factorization to simultaneously learn a consensus clustering indicator matrix and view-specific feature selection matrices. C²IMUFS [7] employs sample information from other views to perform weighted imputation, converting incomplete data into a complete form. Subsequently, it incorporates cross-view complementary and consistency information into an extended non-negative matrix factorization model to identify representative features. Yang et al. [30] proposed a MUFS method that embeds available samples into a low-dimensional space while imposing a low-redundancy constraint to select a non-redundant and informative feature subset. However, these MUFS methods first impute missing data and then perform feature selection, treating the two processes as independent steps and ignoring their potential interactions. Recently, several unified learning-based methods have been proposed to simultaneously perform feature selection and data imputation. Yang et al. [17] exploits low-rank representation and sample diversity learning to recover missing values, and employs graph learning to capture the local structure of the data for identifying discriminative features. TIME-FS [16] integrates adaptive missing data imputation, feature selection, and tensor-based anchor graph learning, enabling mutual reinforcement among these processes. Table I summarizes the comparison of existing multi-view unsupervised feature selection methods. As shown in Table I, most of these methods utilize graph learning to preserve the local manifold structure of the data, thereby aiming to enhance feature selection performance. However, these methods typically rely on similarity graphs based on Euclidean distance to preserve the local manifold

structure, neglecting the topological correlations within the data. Consequently, they fail to accurately capture the intrinsic relationships within the data, resulting in suboptimal feature selection performance. To address this issue, we propose an incomplete multi-view unsupervised feature selection method that leverages consensus topological relationships across views, together with view-specific similarity graphs, to effectively capture the intrinsic data structure.

III. THE PROPOSED METHOD

In this section, we begin by summarizing the notations and definitions utilized throughout the paper. Following this, we introduce the proposed method ATSL-IMUFS for selecting features from incomplete multi-view unlabeled data. An alternative iterative algorithm is then developed to solve the resulting optimization problem. Finally, we analyze the convergence behavior and time complexity of the proposed method.

A. Notations and Definitions

In this paper, all matrices are written in boldface uppercase letters. Given a matrix $\mathbf{M}$, $\mathbf{M}_i$ and $\mathbf{M}_j$ denote the i -th row and the j -th column of $\mathbf{M}$, respectively. M_{ij} represents the element in the i -th row and j -th column of $\mathbf{M}$. Let $\mathbf{M}^T$ denote the transpose of $\mathbf{M}$, and $\text{Tr}(\mathbf{M})$ denote its trace. The Frobenius norm of $\mathbf{M}$ is defined as $\|\mathbf{M}\|_F = \sqrt{\sum_i \sum_j M_{ij}^2}$. The $\ell_{2,1}$ -norm of $\mathbf{M}$ is given by $\|\mathbf{M}\|_{2,1} = \sum_i \sqrt{\sum_j M_{ij}^2}$. Let $\|\mathbf{M}\|_0$ denote the ℓ_0 -norm of $\mathbf{M}$, representing the number of non-zero elements in $\mathbf{M}$. Let $\mathbf{I}$ be the identity matrix, and let $\mathbf{1}$ denote a column vector with all elements equal to 1.

Let $\mathcal{X} = \{\hat{\mathbf{X}}^{(v)}, v = 1, \dots, l\}$ denote an incomplete multi-view dataset consisting of l views, where each data matrix $\hat{\mathbf{X}}^{(v)} \in \mathbb{R}^{d_v \times n}$ comprises d_v features and n instances, with missing entries represented by NaN. Assuming there are n_v missing instances in the v -th view, an indicator matrix $\mathbf{E}^{(v)} \in \mathbb{R}^{n_v \times n}$ is defined as follows to map the missing instances onto the original data matrix:

$$\mathbf{E}_{ij}^{(v)} = \begin{cases} 1, & \text{if the } i\text{-th missing instance is } \hat{\mathbf{X}}_{\cdot j}^{(v)} \text{ in the } v\text{-th view} \\ 0, & \text{otherwise.} \end{cases} \quad (1)$$

Our goal is to select the **top m** most informative features from the incomplete multi-view dataset $\mathcal{X}$.

B. Formulation of ATSL-IMUFS

Spectral graph theory [34] has been widely applied to single-view-based unsupervised feature selection [35] and can



be further extended to address multi-view data, as formulated below:

$$\min_{\mathbf{W}^{(v)}} \sum_{v=1}^l \sum_{i,j=1}^n \|\mathbf{W}^{(v)T} \mathbf{X}_{\cdot i}^{(v)} - \mathbf{W}^{(v)T} \mathbf{X}_{\cdot j}^{(v)}\|_2^2 \mathbf{G}_{ij}^{(v)} + \eta \|\mathbf{W}^{(v)}\|_{2,1}, \quad (2)$$

where $\mathbf{W}^{(v)} \in \mathbb{R}^{d_v \times c}$ is the projection matrix of the v -th view that maps high-dimensional data into a low-dimensional space, $\mathbf{G}^{(v)} \in \mathbb{R}^{n \times n}$ is a similarity-induced graph matrix that captures the pairwise similarity between high-dimensional samples, and η is a regularization parameter that controls the sparsity of $\mathbf{W}^{(v)}$. The first term in Eq. (2) ensures that similar samples in the high-dimensional space remain similar in the low-dimensional space, aiding in effective feature selection. The second term promotes sparsity in the projection matrix $\mathbf{W}^{(v)}$, thereby facilitating the selection of essential features and the exclusion of redundant ones. However, Eq. (2) cannot be directly applied to incomplete multi-view data, as it requires missing values to be imputed with fixed values (such as means) prior to feature selection. This process fails to account for the interaction between data imputation and feature selection. Additionally, the predefined similarity graph in Eq. (2) is easily affected by noise or outliers [36]. Moreover, Eq. (2) does not take into account the varying importance of different views. To address these limitations, we propose a unified model that jointly performs multi-view feature selection and missing data imputation, equipped with adaptive similarity graph learning and view-adaptive weighting, as formulated below:

$$\begin{aligned} \min_{\mathbf{W}^{(v)}, \mathbf{G}^{(v)}, \mathbf{M}^{(v)}, \alpha^{(v)}} & \sum_{v=1}^l [\sum_{i,j=1}^n (\alpha^{(v)})^r \|\mathbf{W}^{(v)T} \mathbf{X}_{\cdot i}^{(v)} - \mathbf{W}^{(v)T} \mathbf{X}_{\cdot j}^{(v)}\|_2^2 \mathbf{G}_{ij}^{(v)} \\ & + \lambda^{(v)} \|\mathbf{G}^{(v)}\|_F^2 + \eta \|\mathbf{W}^{(v)}\|_{2,1}] \\ \text{s.t. } & \mathbf{X}^{(v)} = \mathbf{X}_0^{(v)} + \mathbf{M}^{(v)} \mathbf{E}^{(v)}, \mathbf{W}^{(v)T} \mathbf{W}^{(v)} = \mathbf{I}, \\ & \|\mathbf{G}_{\cdot i}^{(v)}\|_0 = k, \mathbf{G}_{\cdot i}^{(v)} \cdot \mathbf{1} = 1, \mathbf{G}_{ij}^{(v)} \geq 0 (i \neq j), \\ & \mathbf{G}_{ii}^{(v)} = 0, \sum_{v=1}^l \alpha^{(v)} = 1, \alpha^{(v)} \geq 0, \end{aligned} \quad (3)$$

where $\alpha^{(v)}$ denotes the weight of the v -th view, $\mathbf{X}_0^{(v)} \in \mathbb{R}^{d_v \times n}$ is the observable data matrix obtained from the original incomplete data, with the missing entries filled with zeros. $\mathbf{M}^{(v)} \in \mathbb{R}^{d_v \times n_v}$ denotes the imputed matrix for the missing samples in the v -th view, and $\mathbf{X}^{(v)} \in \mathbb{R}^{d_v \times n}$ denotes the complete data matrix, which combines both the **observed entries and the imputed values**. Moreover, orthogonality is imposed on the feature selection matrix $\mathbf{W}^{(v)}$ to ensure that the selected features are uncorrelated [7]. The similarity graph matrix $\mathbf{G}^{(v)}$ is constructed by connecting each sample to its k nearest neighbors, while the Frobenius norm is applied to $\mathbf{G}^{(v)}$ to prevent trivial solutions. $\lambda^{(v)}$ is a trade-off parameter that can be dynamically adjusted, r and η are regularization parameters. In Eq. (3), multi-view feature selection and the adaptive imputation of missing data are seamlessly integrated into a unified learning framework, rather than being handled as separate steps. Moreover, instead of using a predefined similarity graph matrix as in Eq. (2), it is learned adaptively

to reduce the influence of noise and outliers. Finally, an adaptive weighting strategy is employed to capture the relative importance of each view.

Previous studies have confirmed that preserving the local manifold structure is crucial for enhancing the performance of multi-view unsupervised feature selection [11], [7]. In this paper, we also preserve the local manifold structure by learning a consensus topological similarity across different views, which differs from Euclidean-based similarities. In nonlinear manifold spaces, two samples that appear distant in terms of Euclidean distance can still exhibit strong topological consistency if they are connected through a sequence of consecutive neighboring samples. To better capture the intrinsic similarities between samples within the local manifold structure, we propose a consensus topological similarity learning approach as follows:

$$\begin{aligned} \min_{\mathbf{S}, \mathbf{F}} & \sum_{v=1}^l \sum_{i,j,k=1}^n \frac{1}{2} \mathbf{G}_{ij}^{(v)} (\mathbf{S}_{ki} - \mathbf{S}_{kj})^2 + \text{Tr}(\mathbf{F}^T \mathbf{L}_S \mathbf{F}) + \beta \|\mathbf{S}\|_F^2 \\ \text{s.t. } & \mathbf{S}_{\cdot i} \cdot \mathbf{1} = 1, \mathbf{S}_{ij} \geq 0, \mathbf{F}^T \mathbf{F} = \mathbf{I}, \end{aligned} \quad (4)$$

where $\mathbf{S} \in \mathbb{R}^{n \times n}$ denotes the consensus topological similarity matrix across multiple views, $\mathbf{L}_S$ is its corresponding Laplacian matrix, $\mathbf{F} \in \mathbb{R}^{n \times c}$ represents the cluster indicator matrix, and β is a regularization parameter. The first term in Eq. (4) ensures that any two highly similar samples maintain consistent topological relevance with other samples [18], [19], thereby facilitating the propagation of topological connectivity from near to far. The second term in Eq. (4) is formulated to guarantee that samples with high topological similarity are assigned similar cluster labels. By constraining the Frobenius norm of $\mathbf{S}$, the third term ensures that $\mathbf{S}$ does not become a trivial solution.

By combining Eqs. (3) and (4), the proposed incomplete multi-view feature selection method (ATSL-IMUFS) is summarized as follows:

$$\begin{aligned} \min_{\Omega} & \sum_{v=1}^l \sum_{i,j=1}^n (\alpha^{(v)})^r [\sum_{k=1}^n \mathbf{G}_{ij}^{(v)} (\mathbf{S}_{ki} - \mathbf{S}_{kj})^2] + \text{Tr}(\mathbf{F}^T \mathbf{L}_S \mathbf{F}) + \beta \|\mathbf{S}\|_F^2 \\ & + \sum_{v=1}^l [\lambda^{(v)} \|\mathbf{G}^{(v)}\|_F^2 + \eta \|\mathbf{W}^{(v)}\|_{2,1}] \\ \text{s.t. } & \mathbf{X}^{(v)} = \mathbf{X}_0^{(v)} + \mathbf{M}^{(v)} \mathbf{E}^{(v)}, \mathbf{S}_{\cdot i} \cdot \mathbf{1} = 1, \mathbf{S}_{ij} \geq 0, \\ & \|\mathbf{G}_{\cdot i}^{(v)}\|_0 = k, \mathbf{G}_{\cdot i}^{(v)} \cdot \mathbf{1} = 1, \mathbf{G}_{ij}^{(v)} \geq 0 (i \neq j), \mathbf{G}_{ii}^{(v)} = 0, \\ & \mathbf{W}^{(v)T} \mathbf{W}^{(v)} = \mathbf{I}, \mathbf{F}^T \mathbf{F} = \mathbf{I}, \sum_{v=1}^l \alpha^{(v)} = 1, \alpha^{(v)} \geq 0, \end{aligned} \quad (5)$$

$\Omega = \{\mathbf{M}^{(v)}, \mathbf{W}^{(v)}, \mathbf{G}^{(v)}, \alpha^{(v)}, \mathbf{S}, \mathbf{F} | v = 1, \dots, l\}$. By integrating multi-view feature selection, missing data imputation, and local manifold structure learning into a unified framework, Eq. (5) of ATSL-IMUFS enables the exploitation of both intra-view and inter-view local structural information to enhance feature selection and facilitate missing data im-

putation. Furthermore, by simultaneously learning the sample similarity graph for each view and the consensus topological relationship across views, the proposed ATSL-IMUFS more accurately captures the intrinsic similarity within the local manifold structure, which is beneficial for improving feature selection performance.

C. Solution of ATSL-IMUFS

Since the objective function in (5) is not jointly convex with respect to all variables, we propose an alternative iterative algorithm to solve the optimization problem, where each variable is updated sequentially while keeping the others fixed.

1) *Update $\mathbf{M}^{(v)}$ by Fixing Other Variables:* When other variables are fixed, $\mathbf{M}^{(v)}$ can be updated by solving the following optimization problem:

$$\begin{aligned} \min_{\mathbf{M}^{(v)}} \sum_{i,j=1}^n (\alpha^{(v)})^r \|\mathbf{W}^{(v)T} \mathbf{X}_i^{(v)} - \mathbf{W}^{(v)T} \mathbf{X}_j^{(v)}\|_2^2 \mathbf{G}_{ij}^{(v)} \\ \text{s.t. } \mathbf{X}^{(v)} = \mathbf{X}_0^{(v)} + \mathbf{M}^{(v)} \mathbf{E}^{(v)}. \end{aligned} \quad (6)$$

Then, Eq. (6) can be equivalently rewritten in trace form as follows:

$$\begin{aligned} \min_{\mathbf{M}^{(v)}} \text{Tr} [2(\alpha^{(v)})^r \mathbf{W}^{(v)T} (\mathbf{X}_0^{(v)} + \mathbf{M}^{(v)} \mathbf{E}^{(v)}) \mathbf{L}_G^{(v)} (\mathbf{X}_0^{(v)} \\ + \mathbf{M}^{(v)} \mathbf{E}^{(v)})^T \mathbf{W}^{(v)}], \end{aligned} \quad (7)$$

where $\mathbf{L}_G^{(v)}$ is the Laplacian matrix of $\mathbf{G}^{(v)}$. Taking the derivative of Eq. (7) w.r.t $\mathbf{M}^{(v)}$ and setting it to zero yields:

$$\mathbf{M}^{(v)} = -\mathbf{X}_0^{(v)} \mathbf{L}_G^{(v)T} \mathbf{E}^{(v)T} (\mathbf{E}^{(v)} \mathbf{L}_G^{(v)} \mathbf{E}^{(v)T})^{-1}. \quad (8)$$

2) *Update $\mathbf{W}^{(v)}$ by Fixing Other Variables:* By fixing other variables and removing the irrelevant terms, $\mathbf{W}^{(v)}$ can be obtained by solving the following problem:

$$\begin{aligned} \min_{\mathbf{W}^{(v)}} \sum_{i,j=1}^n (\alpha^{(v)})^r \|\mathbf{W}^{(v)T} \mathbf{X}_i^{(v)} - \mathbf{W}^{(v)T} \mathbf{X}_j^{(v)}\|_2^2 \mathbf{G}_{ij}^{(v)} + \eta \|\mathbf{W}^{(v)}\|_{2,1} \\ \text{s.t. } \mathbf{W}^{(v)T} \mathbf{W}^{(v)} = \mathbf{I}. \end{aligned} \quad (9)$$

Eq. (9) can be rewritten in the following equivalent trace form:

$$\min_{\mathbf{W}^{(v)T} \mathbf{W}^{(v)} = \mathbf{I}} \text{Tr}(\mathbf{W}^{(v)T} \mathbf{Z}^{(v)} \mathbf{W}^{(v)}), \quad (10)$$

where $\mathbf{Z}^{(v)} = 2(\alpha^{(v)})^r \mathbf{X}^{(v)} \mathbf{L}_G^{(v)} \mathbf{X}^{(v)T} + \eta \mathbf{D}^{(v)}$ and $\mathbf{D}^{(v)}$ is a diagonal matrix with its i -th element defined as $\mathbf{D}_{ii}^{(v)} = \frac{1}{2\sqrt{\|\mathbf{W}_{i\cdot}^{(v)}\|_2^2 + \varepsilon}}$ (ε is a small constant to avoid division by zero). According to [37], the optimal solution $\mathbf{W}^{(v)}$ in Eq. (10) is derived by selecting the c eigenvectors of $\mathbf{Z}^{(v)}$ corresponding to the c smallest eigenvalues.

3) *Update $\mathbf{G}^{(v)}$ by Fixing Other Variables:* When other variables fixed, solving $\mathbf{G}^{(v)}$ is equal to the following problem:

$$\begin{aligned} \min_{\mathbf{G}^{(v)}} \sum_{i,j=1}^n [\mathbf{U}_{ij}^{(v)} \mathbf{G}_{ij}^{(v)} + \lambda^{(v)} (\mathbf{G}_{ij}^{(v)})^2] \\ \text{s.t. } \|\mathbf{G}_i^{(v)}\|_0 = k, \mathbf{G}_i^{(v)} \cdot \mathbf{1} = 1, \mathbf{G}_{ij}^{(v)} \geq 0 (i \neq j), \mathbf{G}_{ii}^{(v)} = 0, \end{aligned} \quad (11)$$

where $\mathbf{U}_{ij}^{(v)} = (\alpha^{(v)})^r [\|\mathbf{W}^{(v)T} \mathbf{X}_i^{(v)} - \mathbf{W}^{(v)T} \mathbf{X}_j^{(v)}\|_2^2 + \frac{1}{2} \sum_{k=1}^n (\mathbf{S}_{ki} - \mathbf{S}_{kj})^2]$. From Eq. (11), it can be observed that all rows of $\mathbf{G}^{(v)}$ are independent. The i -th row of $\mathbf{G}^{(v)}$ can be updated by solving the following equivalent minimization problem:

$$\begin{aligned} \min_{\mathbf{G}_i^{(v)}} \frac{1}{2} \|\mathbf{G}_i^{(v)}\|_2^2 + \frac{\mathbf{U}_{i\cdot}^{(v)}}{2\lambda^{(v)}} \|\mathbf{G}_i^{(v)}\|_2^2 \\ \text{s.t. } \|\mathbf{G}_i^{(v)}\|_0 = k, \mathbf{G}_i^{(v)} \cdot \mathbf{1} = 1, \mathbf{G}_{ij}^{(v)} \geq 0 (i \neq j), \mathbf{G}_{ii}^{(v)} = 0. \end{aligned} \quad (12)$$

Using the same derivation process as in [36], the optimal solution for $\mathbf{G}_{ij}^{(v)}$ can be obtained as follows:

$$\mathbf{G}_{ij}^{(v)} = \begin{cases} \frac{\mathbf{U}_{i,k+1}^{(v)} - \mathbf{U}_{ij}^{(v)}}{k\mathbf{U}_{i,k+1}^{(v)} - \sum_{j=1}^k \mathbf{U}_{ij}^{(v)}}, j \leq k, \\ 0, j > k. \end{cases} \quad (13)$$

Besides, the trade-off parameter $\lambda^{(v)}$ is calculated as $(k\mathbf{U}_{i,k+1}^{(v)} - \sum_{j=1}^k \mathbf{U}_{ij}^{(v)})/2$ to ensure that $\mathbf{G}_i$ contains k nonzero elements.

4) *Update $\mathbf{S}$ by Fixing Other Variables:* With other variables fixed, the optimization problem for $\mathbf{S}$ can be reformulated as follows:

$$\begin{aligned} \min_{\mathbf{S}} \frac{1}{2} \sum_{v=1}^l (\alpha^{(v)})^r \sum_{i,j,k=1}^n \mathbf{G}_{ij}^{(v)} (\mathbf{S}_{ki} - \mathbf{S}_{kj})^2 + \text{Tr}(\mathbf{F}^T \mathbf{L}_S \mathbf{F}) + \beta \|\mathbf{S}\|_F^2 \\ \text{s.t. } \mathbf{S}_i \cdot \mathbf{1} = 1, \mathbf{S}_{ij} \geq 0. \end{aligned} \quad (14)$$

Since the rows of $\mathbf{S}$ are independent in Eq. (14), each $\mathbf{S}_i$ can be solved individually, as follows:

$$\min_{\mathbf{S}_i \cdot \mathbf{1} = 1, \mathbf{S}_{ij} \geq 0} \mathbf{S}_i \mathbf{A} \mathbf{S}_i^T + \mathbf{B}_i \mathbf{S}_i^T, \quad (15)$$

where $\mathbf{A} = \sum_{v=1}^l (\alpha^{(v)})^r (\mathbf{I} - \mathbf{G}^{(v)}) + \beta \mathbf{I}$ and $\mathbf{B}_i^{(v)} = \frac{1}{2} \sum_{j=1}^n \|\mathbf{F}_i - \mathbf{F}_j\|_2^2$. Following [38], the augmented Lagrangian multiplier (ALM) method is employed to solve Eq. (15). Specifically, by introducing an auxiliary variable $\mathbf{q} \in \mathbb{R}^{1 \times n}$, Eq. (15) can be equivalently transformed into the following optimization problem:

$$\begin{aligned} \min_{\mathbf{q}, \mathbf{S}_i} \mathbf{q} \mathbf{A} \mathbf{S}_i^T + \mathbf{B}_i \mathbf{S}_i^T + \frac{\delta}{2} \|\mathbf{S}_i - \mathbf{q} + \frac{\mathbf{p}}{\delta}\|_2^2 \\ \text{s.t. } \mathbf{S}_i \cdot \mathbf{1} = 1, \mathbf{S}_{ij} \geq 0, \end{aligned} \quad (16)$$

where $\mathbf{p} \in \mathbb{R}^{1 \times n}$ and δ denote the Lagrange multiplier and the penalty parameter, respectively. Subsequently, $\mathbf{q}$, $\mathbf{S}_i$, $\mathbf{p}$ and δ are updated iteratively as follows:

(i) Update of $\mathbf{q}$: The optimization problem with respect to $\mathbf{q}$ is formulated as:

$$\min_{\mathbf{q}} \mathbf{q} \mathbf{A} \mathbf{S}_i^T + \frac{\delta}{2} \|\mathbf{S}_i - \mathbf{q} + \frac{\mathbf{p}}{\delta}\|_2^2. \quad (17)$$

Algorithm 1 Iterative algorithm for solving ATSL-IMUFS

Input: The observed multi-view data matrices $\{\mathbf{X}_0^{(v)}\}_{v=1}^l$, parameters: β , η , and r .

- 1: Initialize: $\mathbf{G}^{(v)}$, $\mathbf{S}$, $\mathbf{D}^{(v)} = \mathbf{I}$ and $\alpha^{(v)} = \frac{1}{l}$, $v = 1, \dots, l$.
- 2: **while** not convergent **do**
- 3: Update $\mathbf{M}^{(v)}$ according to Eq. (8).
- 4: Update $\mathbf{W}^{(v)}$ by solving Eq. (10).
- 5: Update $\mathbf{G}^{(v)}$ according to Eq. (13).
- 6: Update $\mathbf{S}$ by solving Eq. (16).
- 7: Update $\mathbf{F}$ by solving Eq. (23).
- 8: Update $\alpha^{(v)}$ according to Eq. (25).
- 9: **end while**

Output: Sort the ℓ_2 -norm of each row in $\{\mathbf{W}^{(v)}\}_{v=1}^l$ in descending order, and select the top m features.

Taking the derivative of (17) with respect to $\mathbf{q}$, we have

$$\mathbf{q} = \mathbf{S}_i - \frac{1}{\delta}(\mathbf{S}_i \mathbf{A}^T - \mathbf{p}). \quad (18)$$

(ii) Update of $\mathbf{S}_i$: The optimization problem related to $\mathbf{S}_i$ is written as:

$$\begin{aligned} \min_{\mathbf{S}_i} & \|\mathbf{S}_i - \mathbf{q} + \frac{1}{\delta}(\mathbf{p} + \mathbf{q}\mathbf{A} + \mathbf{B}_i)\|_2^2 \\ \text{s.t. } & \mathbf{S}_i \cdot \mathbf{1} = 1, \mathbf{S}_{ij} \geq 0. \end{aligned} \quad (19)$$

Let $\mathbf{J}_i = \mathbf{q} - \frac{1}{\delta}(\mathbf{p} + \mathbf{q}\mathbf{A} + \mathbf{B}_i)$. The Lagrangian function of Eq. (19) is then given by:

$$\min_{\mathbf{S}_i} \|\mathbf{S}_i - \mathbf{J}_i\|_2^2 - \gamma(\mathbf{S}_i \cdot \mathbf{1} - 1) - \mathbf{S}_i \cdot \boldsymbol{\tau}, \quad (20)$$

where γ is a scalar Lagrangian coefficient and $\boldsymbol{\tau}$ is a column vector of Lagrangian coefficients. By applying the Karush–Kuhn–Tucker (KKT) conditions [39], the optimal solution for $\mathbf{S}$ is derived as follows:

$$\mathbf{S}_i = (\mathbf{J}_i - \mathbf{J}_i \mathbf{1} \mathbf{1}^T + \mathbf{1} \mathbf{1}^T, 0)_+. \quad (21)$$

(iii) Update of $\mathbf{p}$ and δ : Following the principles of ALM [40], the Lagrange multiplier $\mathbf{p}$ and the penalty parameter δ are updated as follows:

$$\mathbf{p} = \mathbf{p} + \delta(\mathbf{S}_i - \mathbf{q}), \quad \delta = \delta \times \rho, \quad (22)$$

where $\rho > 1$ is a coefficient that enhances convergence.

5) *Update $\mathbf{F}$ by Fixing Other Variables:* By keeping the other variables fixed, the problem of solving $\mathbf{F}$ is reduced as follows:

$$\min_{\mathbf{F}^T \mathbf{F} = \mathbf{I}} \text{Tr}(\mathbf{F}^T \mathbf{L}_S \mathbf{F}). \quad (23)$$

As stated in [37], the optimal solution for $\mathbf{F}$ is given by the eigenvectors corresponding to the smallest c eigenvalues of $\mathbf{L}_S$.

6) *Update $\alpha^{(v)}$ with Other Variables Fixed:* When the other variables are fixed, $\alpha^{(v)}$ is updated by solving the following problem:

$$\min_{\alpha^{(v)}} \sum_{v=1}^l (\alpha^{(v)})^r h^{(v)}, \quad \text{s.t. } \sum_{v=1}^l \alpha^{(v)} = 1, \alpha^{(v)} \geq 0, \quad (24)$$

where $h^{(v)} = \sum_{i,j=1}^n [\|\mathbf{W}^{(v)T} \mathbf{X}_{\cdot i}^{(v)} - \mathbf{W}^{(v)T} \mathbf{X}_{\cdot j}^{(v)}\|_2^2 \mathbf{G}_{ij}^{(v)} + \frac{1}{2} \sum_{k=1}^n \mathbf{G}_{ij}^{(v)} (\mathbf{S}_{ki} - \mathbf{S}_{kj})^2]$. According to [41], the optimal solution of $\alpha^{(v)}$ can be obtained as follows:

$$\alpha^{(v)} = \frac{h^{(v)^{\frac{1}{1-r}}}}{\sum_{v=1}^l (h^{(v)})^{\frac{1}{1-r}}}. \quad (25)$$

The optimization procedure of ATSL-IMUFS is summarized in Algorithm 1. During the initialization phase, each view-specific matrix $\mathbf{G}^{(v)}$ is initialized using a k -nearest neighbor graph [42], and the consensus matrix $\mathbf{S}$ is initialized as the average of the $\mathbf{G}^{(v)}$ matrices across all views, that is, $\mathbf{S} = \frac{1}{l} \sum_{v=1}^l \mathbf{G}^{(v)}$. The diagonal matrices $\mathbf{D}^{(v)}$ are initialized to identity matrices, and the weights $\alpha^{(v)}$ are set to $\frac{1}{l}$, with l denoting the number of views. The iterative optimization procedure terminates when the relative change in the objective function falls below 10^{-3} .

D. Convergence and Complexity Analysis

In this section, we analyze the convergence and complexity of the proposed ATSL-IMUFS algorithm.

1) *Convergence Analysis:* Since the proposed objective function in Eq. (5) is non-convex with respect to all variables, we address it using the alternating optimization method outlined in Algorithm 1. The convergence of Algorithm 1 is established by demonstrating the convergence of each sub-problem. To this end, we first present the following theorem to prove the convergence of step 4 in Algorithm 1, where $\mathbf{W}^{(v)}$ is updated while keeping the other variables fixed.

Theorem 1. *The update of $\mathbf{W}^{(v)}$ guarantees that the objective function in Eq. (10) monotonically decreases until convergence.*

We provide the detailed convergence proof of Theorem 1 in the supplementary material due to space limitations. Furthermore, in steps 3, 5, and 8 of Algorithm 1, the convergence of the updates for the variables $\mathbf{M}^{(v)}$, $\mathbf{G}^{(v)}$, and $\alpha^{(v)}$ is ensured by their closed-form solutions provided in Eqs. (8), (13), and (27). Regarding step 6 of Algorithm 1, $\mathbf{S}$ is iteratively updated via the augmented Lagrangian method, whose convergence is guaranteed as shown in [40]. In step 7 of Algorithm 1 for updating $\mathbf{F}$, the objective value of Eq. (23) is ensured to decrease monotonically, as demonstrated in [37]. We further empirically validate the convergence behavior of Algorithm 1 in the experimental section.

2) *Complexity Analysis:* Algorithm 1 involves updating six variables. For updating $\mathbf{M}^{(v)}$, the computational cost is primarily determined by Eq. (8), with a complexity of $O(d_v n_v n)$. The update of $\mathbf{W}^{(v)}$ in Eq. (10) requires eigen-decomposition of the $d_v \times d_v$ matrix $\mathbf{Z}^{(v)}$, which incurs a computational cost of $O(d_v^3)$. Since only c eigenvectors associated with the smallest c eigenvalues are needed, we employ the Lanczos method [43] to solve Eq. (10), thereby reducing the computational complexity to $O(d_v^2 c)$, where $c \ll d_v$.

As to updating $\mathbf{G}^{(v)}$, the computational cost consists only of element-wise operations, which can be ignored. Updating $\mathbf{S}$ in Eq. (21) has a computational complexity of $O(n^2)$. According to [36], the computational complexity of updating $\mathbf{F}$ is $O(n^2c)$. Solving α^v involves only numerical division and addition operations, which have a linear computational complexity. Thus, the overall computational complexity of ATSL-IMUFS is $O(T(\sum_{v=1}^l (d_v n_v n + d_v^2 c) + n^2 c))$, where T is the number of iterations.

IV. EXPERIMENTS

In this section, we conduct extensive experiments on several real-world datasets to evaluate the performance of ATSL-IMUFS in comparison with other competing methods.

A. Datasets

In this work, eight real-world multi-view datasets are utilized in our experiments. They are:

Yale¹ consists of 165 grayscale images of 15 subjects, with each subject represented by 11 images capturing different facial expressions.

3Sources² contains 169 news articles collected from three major online news platforms: BBC, Reuters, and The Guardian.

MSRC-v5 [44] is composed of 210 images across 7 object categories, with each image represented by five complementary views.

USPS-MNIST³ consists of handwritten digits across 10 classes, with each digit characterized by both a 784-dimensional MNIST image and a 256-dimensional USPS image.

MSRA-6view [45] is composed of 210 object images, each depicted through six distinct views: CENT, CMT, GIST, HOG, LBP, and SIFT.

100leaves⁴ contains 1,600 binary leaf images from 100 plant species, with each image described by three views: shape, margin, and texture.

LandUse-21⁵ consists of 2,100 satellite images captured by Landsat, spanning 21 distinct land-use classes.

SensIT Vehicle [46] contains 30,000 samples collected from a wireless sensor network, with each sample comprising signals from both acoustic and seismic sensors. Following [47], a subset of 3,000 samples is constructed by randomly selecting 1,000 samples from each class.

The detailed statistics of these multi-view datasets are provided in Table II. In addition, we simulate the incomplete multi-view setting following the method proposed in [7], by randomly removing specific proportions of samples from each view as missing data. The missing rates vary from 10% to 50% in 10% increments.

TABLE II: A detailed description of datasets

Datasets	Abbr.	# Instances	# Classes	# Views	# Features
Yale	Yale	165	15	4	256/256/256/256
3Sources	3S	169	6	3	3560/3631/3068
MSRC-v5	MSRC	210	7	5	24/576/512/256/254
USPS-MNIST	USPS	500	10	2	784/256
MSRA-6view	MSRA	210	7	6	1302/48/512/100/256/210
100leaves	100L	1600	100	3	64/64/64
LandUse-21	Land	2100	21	3	20/59/40
SensIT Vehicle	Sens	3000	3	2	50/50

B. Experimental Setup

Following previous studies [11], [7], [15], we employ the k-means clustering algorithm [48] on the selected features to assess the performance of the proposed ATSL-IMUFS. To evaluate the quality of the selected features produced by different feature selection methods, two commonly used clustering metrics, clustering accuracy (ACC) and normalized mutual information (NMI), are utilized, which are defined as follows:

$$\text{ACC} = \frac{1}{n} \sum_{i=1}^n \delta(y_i, \text{map}(y'_i)), \quad (26)$$

where n denotes the total number of samples, and y_i and y'_i represent the ground-truth and predicted cluster labels of the i -th sample, respectively. The function $\text{map}(\cdot)$ assigns each cluster to its optimal true label using the Kuhn-Munkres algorithm [49], while $\delta(x, y)$ is an indicator function that equals 1 if $x = y$ and 0 otherwise.

Let $\mathcal{C}$ and $\mathcal{C}'$ denote the ground truth cluster set and the predicted cluster set, respectively. NMI is formally defined as follows:

$$\text{NMI} = \frac{MI(\mathcal{C}, \mathcal{C}')}{\max(H(\mathcal{C}), H(\mathcal{C}'))}, \quad (27)$$

where $MI(\mathcal{C}, \mathcal{C}')$ denotes the mutual information between $\mathcal{C}$ and $\mathcal{C}'$, while $H(\mathcal{C})$ and $H(\mathcal{C}')$ denote the entropies of $\mathcal{C}$ and $\mathcal{C}'$, respectively. For the two aforementioned metrics, higher values indicate better performance.

Meanwhile, the proposed ATSL-IMUFS is compared against several competing approaches, including both single-view and multi-view unsupervised feature selection methods, which are briefly introduced below.

- **AllFea** combines features from different views to be compared.
- **LS** [8] is a classical single-view unsupervised feature selection method that uses the Laplacian Score to select features.
- **HSL** [10] is a recent single-view unsupervised feature selection method that unifies feature selection with the learning of both first-order and high-order similarities within a single framework.
- **UKMFS** [23] selects important features by utilizing binary hashing codes derived from multiple views.
- **GCDUFS** [24] introduces a unified framework for feature selection that achieves both view consensus and diversity learning simultaneously.

¹<http://www.cad.zju.edu.cn/home/dengcai/Data/FaceData.html>

²<http://mlg.ucd.ie/datasets/3sources.html>

³<https://paperswithcode.com/sota/domain-adaptation-on-mnist-to-usps>

⁴<https://archive.ics.uci.edu/dataset/241>

⁵<https://www.kaggle.com/datasets/apollo2506/landuse-scene-classification>

TABLE III: Means and standard deviations (%) of ACC for various methods across different datasets. The best results are highlighted in bold.

Datasets Methods	Yale	3S	MSRC	USPS	MSRA	100L	Land	Sens
ATSL-IMUFS	35.54 ± 2.20	37.18 ± 4.62	74.52 ± 1.75	44.55 ± 2.84	65.57 ± 2.08	42.05 ± 0.93	18.81 ± 0.70	58.01 ± 0.02
AllFea	25.54 ± 2.68●	34.10 ± 1.44●	42.60 ± 2.83●	34.67 ± 3.02●	42.35 ± 3.47●	31.22 ± 1.06●	14.19 ± 0.93●	46.67 ± 0.03●
LS	24.79 ± 2.16●	34.00 ± 1.62●	41.00 ± 1.67●	34.33 ± 3.66●	43.43 ± 3.85●	25.81 ± 0.92●	14.53 ± 0.99●	46.70 ± 0.00●
HSL	25.45 ± 2.50●	34.34 ± 2.36●	41.43 ± 1.90●	35.03 ± 2.87●	40.94 ± 3.63●	30.80 ± 0.71●	14.88 ± 0.98●	46.70 ± 0.04●
UKMFS	24.36 ± 2.24●	34.71 ± 3.93●	35.67 ± 3.05●	35.31 ± 3.09●	45.98 ± 5.24●	31.08 ± 0.89●	13.04 ± 0.64●	46.45 ± 1.23●
GCDUFS	23.72 ± 2.40●	33.65 ± 1.20●	40.16 ± 3.45●	33.87 ± 2.70●	48.75 ± 4.22●	29.83 ± 0.84●	12.63 ± 0.40●	40.39 ± 0.71●
CFSMO	24.40 ± 1.69●	34.73 ± 3.14●	42.75 ± 2.92●	36.02 ± 2.99●	38.57 ± 3.27●	31.29 ± 0.70●	14.74 ± 1.13●	46.64 ± 3.22●
CE-UMFS	25.68 ± 2.56●	34.50 ± 1.77●	44.05 ± 2.40●	37.10 ± 2.87●	42.94 ± 4.39●	29.29 ± 0.89●	15.24 ± 0.50●	42.94 ± 1.76●
SCMvFS	24.75 ± 1.79●	35.44 ± 2.93○	43.16 ± 2.73●	35.29 ± 3.44●	49.83 ± 5.05●	26.02 ± 0.76●	15.39 ± 1.04●	45.38 ± 0.07●
TRCA-CGL	24.71 ± 1.81●	35.44 ± 3.39○	44.21 ± 3.35●	35.74 ± 2.98●	45.63 ± 4.32●	32.53 ± 0.86●	15.23 ± 0.57●	47.85 ± 1.54●
SDFS	25.80 ± 2.19●	34.91 ± 2.16●	41.30 ± 1.87●	33.05 ± 2.06●	38.94 ± 3.76●	30.15 ± 0.77●	14.07 ± 0.87●	46.71 ± 0.03●
JMVFG	27.68 ± 2.56●	35.46 ± 3.00○	45.57 ± 3.67●	36.80 ± 2.02●	50.27 ± 5.43●	31.50 ± 0.70●	15.03 ± 0.91●	47.62 ± 0.02●
TLR-MUFS	25.37 ± 2.27●	34.58 ± 3.50●	45.35 ± 3.50●	35.64 ± 2.00●	46.57 ± 3.83●	30.01 ± 0.77●	12.95 ± 0.59●	40.75 ± 0.22●
CVFS	24.97 ± 1.85●	35.19 ± 3.02●	43.49 ± 2.37●	36.18 ± 2.37●	48.06 ± 3.31●	27.48 ± 0.66●	14.88 ± 1.04●	47.10 ± 0.05●
C ² IMUFS	24.36 ± 1.72●	35.46 ± 3.75○	42.48 ± 2.07●	37.01 ± 2.47●	39.48 ± 3.06●	27.84 ± 0.86●	15.00 ± 0.96●	46.83 ± 0.65●
UNIFIER	26.06 ± 1.69●	38.22 ± 4.36	44.89 ± 4.38●	37.75 ± 2.80●	55.06 ± 4.37●	31.79 ± 0.81●	14.19 ± 0.59●	48.63 ± 0.02●
TERUIMUFS	20.20 ± 0.85●	35.60 ± 3.25○	25.97 ± 1.99●	28.49 ± 1.20●	23.24 ± 0.72●	30.00 ± 0.88●	11.97 ± 0.59●	34.53 ± 0.00●
UIMUFSLR	21.43 ± 1.26●	34.91 ± 2.56●	31.60 ± 1.22●	35.08 ± 1.97●	38.56 ± 2.41●	27.06 ± 0.69●	10.29 ± 0.31●	34.53 ± 0.00●
TIME-FS	24.10 ± 1.59●	34.67 ± 3.31●	42.76 ± 2.32●	36.52 ± 2.91●	38.60 ± 3.07●	27.68 ± 0.74●	14.96 ± 0.81●	46.71 ± 0.11●

- **CFSMO** [25] leverages multi-order neighborhood information derived from similarity graphs to perform feature selection.
- **CE-UMFS** [26] selects features by utilizing the Hilbert–Schmidt independence criterion to preserve view exclusivity and the nuclear norm to enforce inter-view consistency.
- **SCMvFS** [27] selects discriminative features by utilizing both complementary heterogeneous and homogeneous information from multiple views.
- **TRCA-CGL** [28] combines adaptive consensus similarity learning with tensor robust principal component analysis in a unified framework for feature selection.
- **SDFS** [29] constructs a consensus similarity graph by integrating weighted view-specific similarity graphs to enhance feature selection.
- **JMVFG** [11] simultaneously learns cluster assignments, captures both global and local similarity structures, and accounts for multi-view consistency and inconsistency during feature selection.
- **TLR-MUFS** [12] constructs view-specific graphs to exploit complementary information and enforces a tensor low-rank constraint for cross-view consistency in feature selection.
- **CVFS** [14] utilizes instance-level and feature-level interactions across views to select discriminative features from incomplete multi-view data.
- **C²IMUFS** [7] introduces an incomplete multi-view unsupervised feature selection method that simultaneously leverages complementary and consensus information across views.

- **UNIFIER** [15] combines missing data imputation, feature selection, and adaptive similarity graph learning within a unified framework.
- **TERUIMUFS** [17] leverages self-representation learning with a low-rank tensor constraint to capture the intrinsic structure of the data and identify discriminative features.
- **UIMUFSLR** [30] embeds multi-view data into a shared low-dimensional space and performs feature selection with an explicit low-redundancy constraint to obtain a discriminative and non-redundant feature subset.
- **TIME-FS** [16] jointly learns a consistent anchor graph and a view-preference weight matrix, thereby facilitating missing data imputation and feature selection.

The aforementioned single-view-based methods conduct feature selection on concatenated data from different views. Besides, since most comparison methods are not directly applicable to incomplete multi-view data, the missing samples are initially imputed with the mean value of their corresponding view prior to employing these methods. Furthermore, the parameters of the compared methods are tuned using grid search to achieve optimal results. Similarly, we employ grid search to optimize the parameters of our method, where β and η are selected from $\{10^{-3}, 10^{-2}, 10^{-1}, 1, 10, 10^2, 10^3\}$, and r is chosen from $\{2, 3, 4, 5, 6\}$. According to [50] and [51], parallel computing can be utilized to enhance the efficiency of optimal parameter grid search. Following common practice in previous literature [42], we set $k = 15$ in our method. Since determining the optimal number of selected features for a given dataset remains a challenging problem [4], we vary the percentage of selected features from 10% to 50% in steps of 10% across all datasets. Then, k-means clustering

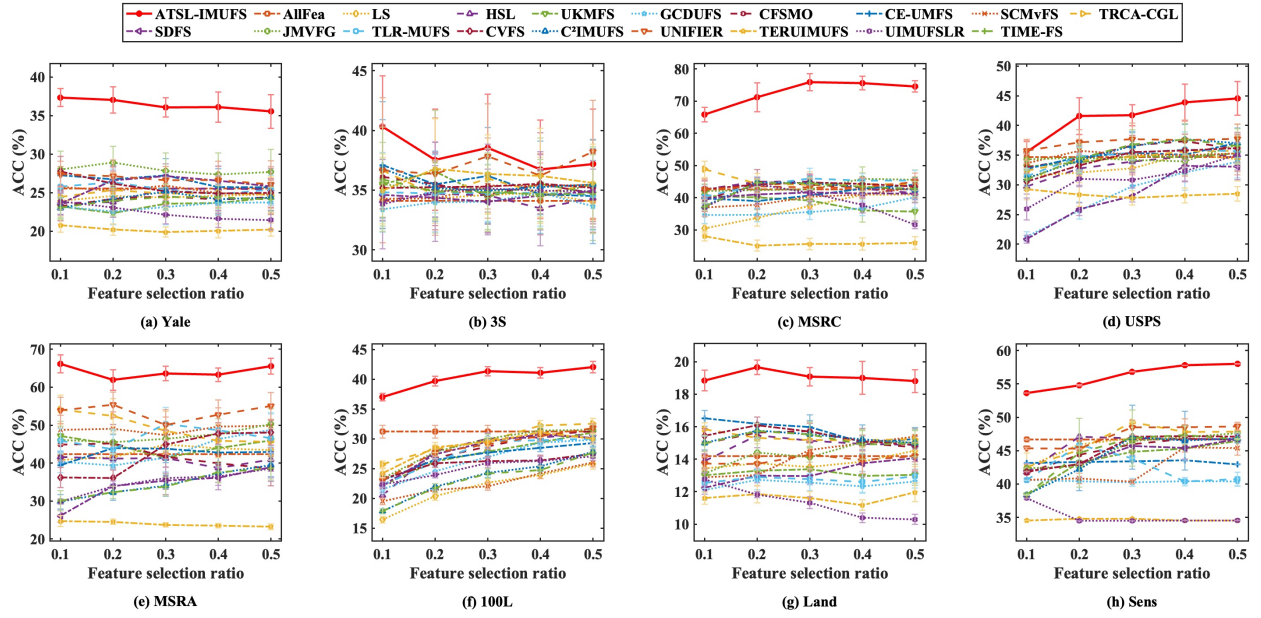


Fig. 2: ACC of different methods under varying feature selection ratios.

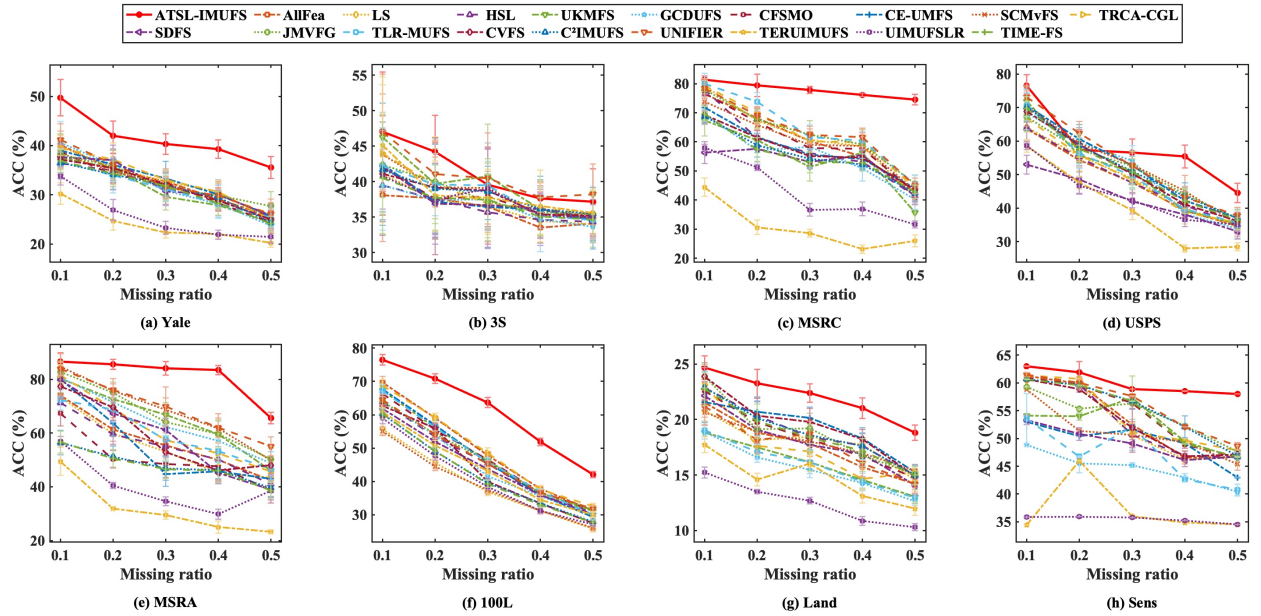


Fig. 3: ACC of different methods under varying missing ratios.

is performed 30 times on the selected features, and the average results along with the standard deviation are reported. The implementation of our method is publicly available at <https://github.com/Lin-Xu00/ATSL-IMUFS>.

C. Experimental Results

Table III summarizes the experimental results for ACC under a 50% missing rate and a 50% feature selection rate. Additionally, the Wilcoxon rank-sum test [52] is employed to verify whether the proposed ATSL-IMUFS significantly outperforms the other methods. The values marked with ●/○ in Table III indicate that the proposed method is statistically

superior or inferior to the other methods at a significance level of 0.05. Owing to space limitations, the NMI clustering results of different feature selection methods are presented in Table I of the supplementary material.

As shown in Table III and Table I in the supplementary material, the proposed ATSL-IMUFS outperforms other methods in most cases. As to MSRC dataset, ATSL-IMUFS achieves an average improvement of over 32% in both ACC and NMI compared to other competing methods. As to Yale, MSRA, 100L, and Sens datasets, ATSL-IMUFS gains over 14% in ACC and 11% in NMI on average improvement compared to all other methods. For USPS and Land, ATSL-IMUFS

also obtains more than 6% improvement in average ACC and NMI. On the 3S dataset, ATSL-IMUFS achieves the second-best performance in ACC and outperforms all other methods in NMI. Additionally, ATSL-IMUFS consistently outperforms the baseline method AllFea in both ACC and NMI, demonstrating the effectiveness of the proposed method. Furthermore, ATSL-IMUFS outperforms all single-view-based feature selection methods, achieving average improvements of more than 10% in both ACC and NMI across all datasets. This demonstrates the effectiveness of our method in utilizing the intra-view similarity graph and capturing the consensus topological relationships across views, compared to single-view methods that simply stack all views together.

Given the difficulty in determining the optimal number of selected features for different datasets, we also provide a performance comparison of various methods under varying feature selection ratios. Due to space limitations, only the ACC results are shown here, while the NMI results can be found in Fig. 1 of the supplementary material. Fig. 2 presents the ACC values for the compared methods at a fixed missing ratio of 50% and varying feature selection ratios across different datasets, with all reported results shown with error bars. As can be seen from these results, the proposed ATSL-IMUFS outperforms other methods across a range of feature selection ratios in most cases. Furthermore, we evaluate the performance of the compared methods in terms of ACC and NMI under varying missing ratios, with the feature selection ratio fixed at 50%. The NMI results for different missing ratios are provided in Fig. 2 of the supplementary material due to space limitations. Fig. 3 shows the ACC results across different datasets as the missing ratio varies from 10% to 50%, with all reported results including error bars. As observed in these figures, the proposed method demonstrates superior performance compared to other methods in most cases. The superior performance of ATSL-IMUFS is attributed to its unified framework, which integrates multi-view feature selection, missing data imputation, and local manifold learning capable of exploring both intra-view and inter-view similarity structures, thereby enabling these components to enhance one another synergistically.

D. Parameter Sensitivity

The proposed objective function in Eq. (5) involves four parameters: β , η , r , and $\lambda^{(v)}$, where $\lambda^{(v)}$ is automatically determined during the solution of $\mathbf{G}^{(v)}$ in Eq. (13). To investigate the impact of these three parameters (i.e., β , η , r) on the proposed ATSL-IMUFS, we show its performance by varying one parameter while keeping the other two fixed. Fig. 4 shows the ACC values of ATSL-IMUFS for different values of β , η , r , and feature selection ratios (FR). Due to space constraints, the corresponding NMI results are presented in Fig. 3 of the supplementary material. We observe that the parameters r and η are somewhat sensitive in our method with respect to both ACC and NMI. In comparison, the parameter β is even more sensitive than r and η . We need to tune the parameter β using a grid search method in practice to achieve optimal performance.

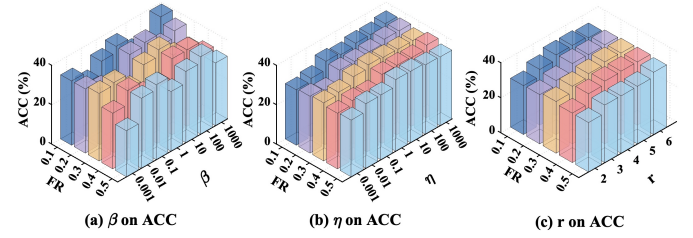


Fig. 4: ACC of ATSL-IMUFS with varying parameters β , η , r and feature selection ratios on 100L dataset.

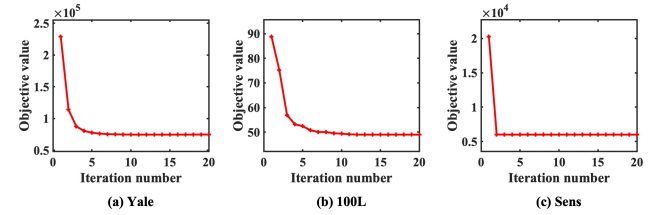


Fig. 5: Convergence curves of ATSL-IMUFS on Yale, 100L and Sens datasets.

E. Convergence Behavior

In this section, we evaluate the convergence speed of our method empirically. Fig. 5 shows that the objective values of Eq. (5) change with the number of iterations on Yale, 100L and Sens datasets. The convergence curves of ATSL-IMUFS on other five datasets (3S, MSRC, USPS, MSRA and Land) are provided in Fig. 4 of the supplementary material. From these figures, we observe that the convergence curve of the proposed algorithm drops sharply during the initial iterations and stabilizes at approximately 15 iterations.

F. Ablation Study

In this section, we conduct ablation experiments to demonstrate the contributions of individual components in the proposed ATSL-IMUFS. To this end, we designed three variants of ATSL-IMUFS: ATSL-IMUFS-I removes the adaptive imputation module and uses mean imputation for missing values; ATSL-IMUFS-II replaces the adaptive similarity graph learning module with pre-defined graphs constructed via the Gaussian kernel method described in [10]; and ATSL-IMUFS-III removes the consensus topological similarity learning module.

The results of the ablation experiments are presented in Table IV. It can be seen that ATSL-IMUFS consistently outperforms its three variants across all datasets in terms of both ACC and NMI. Compared to ATSL-IMUFS-I, ATSL-IMUFS achieves significantly better performance, demonstrating the effectiveness of the proposed unified learning framework. Besides, the superiority of ATSL-IMUFS over ATSL-IMUFS-II indicates that adaptive similarity graph learning helps to reduce the influence of noise and outliers. Moreover, ATSL-IMUFS significantly outperforms ATSL-IMUFS-III, demonstrating the effectiveness of the proposed consensus topological similarity learning module in better capturing the intrinsic similarity within the local manifold structure.

TABLE IV: Comparison of clustering performance between ATSL-IMUFS and its three variants across eight datasets in terms of ACC and NMI.

Metrics	Methods	Yale	3S	MSRC	USPS	MSRA	100L	Land	Sens
ACC	ATSL-IMUFS	35.54 ± 2.20	37.18 ± 4.62	74.52 ± 1.75	44.55 ± 2.84	65.57 ± 2.08	42.05 ± 0.93	18.81 ± 0.70	58.01 ± 0.02
	ATSL-IMUFS-I	23.82 ± 1.67	34.85 ± 3.67	30.70 ± 0.66	30.75 ± 1.66	28.87 ± 0.86	23.75 ± 0.80	13.03 ± 0.64	39.38 ± 1.25
	ATSL-IMUFS-II	24.83 ± 1.97	35.72 ± 5.08	34.24 ± 1.75	25.24 ± 0.90	29.51 ± 0.58	24.40 ± 0.69	13.23 ± 0.66	39.15 ± 0.91
	ATSL-IMUFS-III	20.04 ± 0.83	35.60 ± 3.21	22.03 ± 0.33	34.21 ± 1.71	30.02 ± 0.72	23.61 ± 0.65	11.05 ± 0.39	45.38 ± 0.05
NMI	ATSL-IMUFS	38.24 ± 1.60	7.18 ± 2.10	60.55 ± 1.10	35.50 ± 2.09	45.40 ± 1.48	64.85 ± 0.42	18.85 ± 0.54	14.38 ± 0.02
	ATSL-IMUFS-I	27.07 ± 1.72	6.42 ± 3.09	18.86 ± 0.93	22.45 ± 1.06	19.67 ± 0.92	45.89 ± 0.65	13.38 ± 0.65	2.56 ± 0.78
	ATSL-IMUFS-II	27.68 ± 1.62	8.36 ± 4.90	20.50 ± 1.31	19.34 ± 0.74	18.82 ± 0.46	46.14 ± 0.56	13.82 ± 0.73	2.39 ± 0.80
	ATSL-IMUFS-III	23.63 ± 1.09	5.82 ± 2.75	4.63 ± 0.57	25.09 ± 1.32	19.92 ± 1.12	48.94 ± 0.71	8.53 ± 0.49	4.55 ± 0.03

G. Visualization Analysis

To intuitively illustrate the effectiveness of the proposed topological similarity graph in capturing the intrinsic data structure, we visualize the similarity graphs generated by the top three methods on the MSRC dataset, as shown in Fig. 6. In this figure, panel (a) presents our approach, which achieves the best performance, while panels (b) and (c) present the second- and third-best methods, respectively. As can be observed, the topological similarity graph learned by ATSL-IMUFS clearly demonstrates the underlying clustering structure. In contrast, the similarity graphs from the other two methods, which are based on Euclidean distance, lack a discernible clustering structure and do not effectively capture the local structure. These results further demonstrate the superiority of ATSL-IMUFS in revealing the intrinsic data structure.

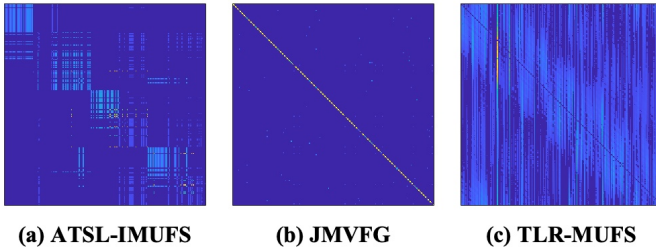


Fig. 6: Visualizations of the similarity graphs learned by the top three methods on MSRC dataset.

To further evaluate the effectiveness of the selected features, we apply t-SNE [53] to project the samples onto a two-dimensional space, with missing instances highlighted by red stars. The visualization results are presented in Figs. 7 and 8 for the MSRC and MSRA datasets, with a 50% feature selection ratio and a 50% missing ratio. In both figures, the top three methods are compared: our method, which achieved the best results, is shown in panel (a), while the second and third best methods are displayed in panels (b) and (c), respectively. As illustrated, the second and third best methods exhibit substantial overlap among clusters, whereas ATSL-IMUFS yields clearer inter-cluster separation with more distinct boundaries. Moreover, compared to these methods, ATSL-IMUFS correctly assigns missing samples to their respective clusters, highlighting its ability to identify compact

and discriminative feature subsets from incomplete multi-view data.

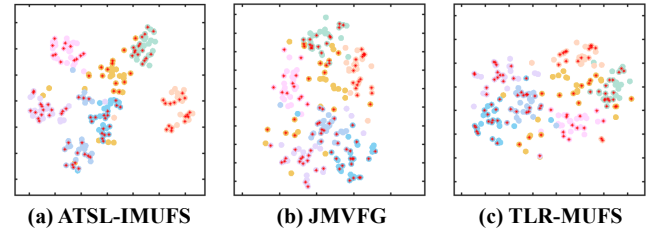


Fig. 7: t-SNE visualizations of the top three methods on MSRC dataset.

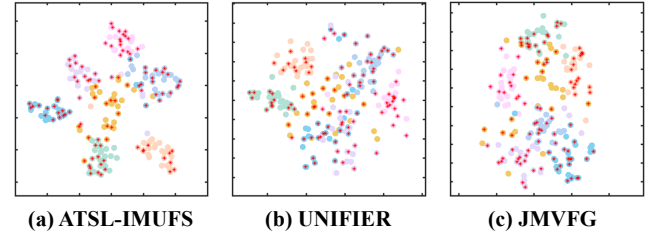


Fig. 8: t-SNE visualizations of the top three methods on MSRA dataset.

V. CONCLUSION AND DISCUSSION

In this paper, we propose a novel MUFS method, named adaptive topological similarity learning for incomplete multi-view unsupervised feature selection (ATSL-IMUFS). This method integrates multi-view feature selection, missing data imputation, and the exploration of both intra-view and inter-view local structures into a unified learning framework, in which these components work together to mutually reinforce one another. Meanwhile, in order to ensure accurate characterization of the local manifold structure, the proposed ATSL-IMUFS adaptively learns consensus topological relationships across views as well as the similarity graphs associated with each view. Extensive experimental results on eight real-world multi-view datasets, along with detailed analyses of parameter sensitivity and convergence behavior, validate the efficacy of the proposed ATSL-IMUFS.

Although the proposed ATSL-IMUFS demonstrates superior performance, it leaves several important questions open for future investigation. Since the objective function is non-convex with respect to multiple variables, the developed alternating iterative algorithm cannot guarantee global convergence, which is a common limitation of existing multi-view unsupervised feature selection methods. In future work, we aim to develop a robust incomplete multi-view unsupervised feature selection method that achieves global convergence and is insensitive to variations in missing data rates. Another limitation is the high computational cost associated with grid search. To address this, we intend to investigate automated hyperparameter optimization techniques to further improve the efficiency of our method. Moreover, our method may be less effective in scenarios with class distribution imbalance, where the learned graph structure could underrepresent minority-class samples. This may lead to insufficient connectivity among minority samples and limit the ability to capture their local manifold structure, ultimately affecting feature selection performance. Addressing these limitations remains an important direction for future research. In many real-world applications, obtaining a large number of labeled samples is often impractical, and usually only a small set is available. Building on this, our future work will leverage the available label information to guide local manifold structure learning, thus extending our proposed unsupervised framework to a semi-supervised setting for feature selection in incomplete multi-view data.

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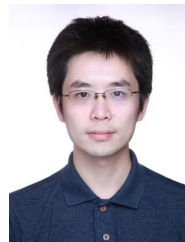
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